

# Generative Adversarial Nets

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## Abstract

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model  $G$  that captures the data distribution, and a discriminative model  $D$  that estimates the probability that a sample came from the training data rather than  $G$ . The training procedure for  $G$  is to maximize the probability of  $D$  making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions  $G$  and  $D$ , a unique solution exists, with  $G$  recovering the training data distribution and  $D$  equal to  $\frac{1}{2}$  everywhere. In the case where  $G$  and  $D$  are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

## 1 Introduction

The promise of deep learning is to discover rich, hierarchical models [2] that represent probability distributions over the kinds of data encountered in artificial intelligence applications, such as natural images, audio waveforms containing speech, and symbols in natural language corpora. So far, the most striking successes in deep learning have involved discriminative models, usually those that map a high-dimensional, rich sensory input to a class label [14, 22]. These striking successes have primarily been based on the backpropagation and dropout algorithms, using piecewise linear units [19, 9, 10] which have a particularly well-behaved gradient. Deep *generative* models have had less of an impact, due to the difficulty of approximating many intractable probabilistic computations that arise in maximum likelihood estimation and related strategies, and due to difficulty of leveraging the benefits of piecewise linear units in the generative context. We propose a new generative model estimation procedure that sidesteps these difficulties.<sup>1</sup>

In the proposed *adversarial nets* framework, the generative model is pitted against an adversary: a discriminative model that learns to determine whether a sample is from the model distribution or the data distribution. The generative model can be thought of as analogous to a team of counterfeiters, trying to produce fake currency and use it without detection, while the discriminative model is analogous to the police, trying to detect the counterfeit currency. Competition in this game drives both teams to improve their methods until the counterfeits are indistinguishable from the genuine articles.

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<sup>1</sup>All code and hyperparameters available at <http://www.github.com/goodfeli/adversarial>

# 生成对抗网络

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## 摘要

我们提出了一种新的框架，通过对抗过程来估计生成模型。在此框架中，我们同时训练两个模型：一个是捕获数据分布的生成模型 $G$ ，另一个是判别模型 $D$ ，用于估计样本来自训练数据而非 $G$ 的概率。 $G$ 的训练过程旨在最大化 $D$ 犯错的概率。该框架对应一个极小极大二人博弈。在任意函数 $G$ 和 $D$ 的空间中，存在唯一解，其中 $G$ 恢复训练数据分布，且 $D$ 处处等于 $\frac{1}{2}$ 。当 $G$ 和 $D$ 由多层感知器定义时，整个系统可以通过反向传播进行训练。在训练或生成样本过程中，无需任何马尔可夫链或展开的近似推理网络。实验通过对生成样本进行定性和定量评估，证明了该框架的潜力。

## 1 引言

深度学习的承诺在于发现丰富、层次化的模型[2]，这些模型能够表示人工智能应用中遇到的各种数据的概率分布，例如自然图像、包含语音的音频波形以及自然语言语料库中的符号。迄今为止，深度学习最引人注目的成功主要涉及判别模型，通常是那些将高维、丰富的感官输入映射到类别标签的模型[14, 22]。这些显著的成功主要基于反向传播和dropout算法，使用了具有特别良好梯度的分段线性单元[19, 9, 10]。深度*generative*模型的影响较小，这是由于在最大似然估计及相关策略中，近似许多难以处理的概率计算存在困难，以及在生成模型中利用分段线性单元的优势存在挑战。我们提出了一种新的生成模型估计方法，以规避这些困难。<sup>1</sup>

在所提出的*adversarial nets*框架中，生成模型与一个对手——判别模型进行对抗：判别模型学习判断样本是来自模型分布还是数据分布。生成模型可以类比为一组伪造者，试图制造假币并在不被发现的情况下使用，而判别模型则类似于警察，试图检测出假币。这场博弈中的竞争推动双方改进各自的方法，直到伪造品与真品无法区分。

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This framework can yield specific training algorithms for many kinds of model and optimization algorithm. In this article, we explore the special case when the generative model generates samples by passing random noise through a multilayer perceptron, and the discriminative model is also a multilayer perceptron. We refer to this special case as *adversarial nets*. In this case, we can train both models using only the highly successful backpropagation and dropout algorithms [17] and sample from the generative model using only forward propagation. No approximate inference or Markov chains are necessary.

## 2 Related work

An alternative to directed graphical models with latent variables are undirected graphical models with latent variables, such as restricted Boltzmann machines (RBMs) [27, 16], deep Boltzmann machines (DBMs) [26] and their numerous variants. The interactions within such models are represented as the product of unnormalized potential functions, normalized by a global summation/integration over all states of the random variables. This quantity (the *partition function*) and its gradient are intractable for all but the most trivial instances, although they can be estimated by Markov chain Monte Carlo (MCMC) methods. Mixing poses a significant problem for learning algorithms that rely on MCMC [3, 5].

Deep belief networks (DBNs) [16] are hybrid models containing a single undirected layer and several directed layers. While a fast approximate layer-wise training criterion exists, DBNs incur the computational difficulties associated with both undirected and directed models.

Alternative criteria that do not approximate or bound the log-likelihood have also been proposed, such as score matching [18] and noise-contrastive estimation (NCE) [13]. Both of these require the learned probability density to be analytically specified up to a normalization constant. Note that in many interesting generative models with several layers of latent variables (such as DBNs and DBMs), it is not even possible to derive a tractable unnormalized probability density. Some models such as denoising auto-encoders [30] and contractive autoencoders have learning rules very similar to score matching applied to RBMs. In NCE, as in this work, a discriminative training criterion is employed to fit a generative model. However, rather than fitting a separate discriminative model, the generative model itself is used to discriminate generated data from samples a fixed noise distribution. Because NCE uses a fixed noise distribution, learning slows dramatically after the model has learned even an approximately correct distribution over a small subset of the observed variables.

Finally, some techniques do not involve defining a probability distribution explicitly, but rather train a generative machine to draw samples from the desired distribution. This approach has the advantage that such machines can be designed to be trained by back-propagation. Prominent recent work in this area includes the generative stochastic network (GSN) framework [5], which extends generalized denoising auto-encoders [4]: both can be seen as defining a parameterized Markov chain, i.e., one learns the parameters of a machine that performs one step of a generative Markov chain. Compared to GSNs, the adversarial nets framework does not require a Markov chain for sampling. Because adversarial nets do not require feedback loops during generation, they are better able to leverage piecewise linear units [19, 9, 10], which improve the performance of backpropagation but have problems with unbounded activation when used in a feedback loop. More recent examples of training a generative machine by back-propagating into it include recent work on auto-encoding variational Bayes [20] and stochastic backpropagation [24].

## 3 Adversarial nets

The adversarial modeling framework is most straightforward to apply when the models are both multilayer perceptrons. To learn the generator’s distribution  $p_g$  over data  $x$ , we define a prior on input noise variables  $p_z(z)$ , then represent a mapping to data space as  $G(z; \theta_g)$ , where  $G$  is a differentiable function represented by a multilayer perceptron with parameters  $\theta_g$ . We also define a second multilayer perceptron  $D(x; \theta_d)$  that outputs a single scalar.  $D(x)$  represents the probability that  $x$  came from the data rather than  $p_g$ . We train  $D$  to maximize the probability of assigning the correct label to both training examples and samples from  $G$ . We simultaneously train  $G$  to minimize  $\log(1 - D(G(z)))$ :

该框架能够为多种模型和优化算法提供具体的训练算法。在本文中，我们探讨了一种特殊情况：生成模型通过将随机噪声传递到多层感知机来生成样本，而判别模型同样采用多层感知机结构。我们将这种特殊情况称为*adversarial nets*。在此情况下，我们可以仅使用极为成功的反向传播和dropout算法[17]来训练两个模型，并仅通过前向传播从生成模型中采样。无需近似推理或马尔可夫链。

## 2 相关工作

具有潜在变量的有向图模型的替代方案是具有潜在变量的无向图模型，例如受限玻尔兹曼机（RBM）[27, 16]、深度玻尔兹曼机（DBM）[26]及其众多变体。此类模型中的相互作用表示为未归一化势函数的乘积，并通过随机变量所有状态的全局求和/积分进行归一化。这个量（*partition function*）及其梯度对于除最平凡实例外的所有情况都是难以处理的，尽管它们可以通过马尔可夫链蒙特卡罗（MCMC）方法进行估计。对于依赖MCMC的学习算法而言，混合是一个重要问题[3, 5]。

深度信念网络（DBNs）[16]是一种混合模型，包含一个无向层和多个有向层。虽然存在一种快速的逐层近似训练准则，但DBNs同时带来了与无向模型和有向模型相关的计算难题。

除了近似或限制对数似然的方法外，还提出了其他不依赖于此的标准，例如分数匹配[18]和噪声对比估计（NCE）[13]。这两种方法都要求所学习的概率密度在归一化常数范围内能够解析指定。需要注意的是，在许多具有多层潜在变量的有趣生成模型（如DBNs和DBMs）中，甚至无法推导出易于处理的未归一化概率密度。一些模型，如去噪自编码器[30]和收缩自编码器，其学习规则与应用于RBMs的分数匹配非常相似。在NCE中，与本研究一样，采用判别性训练准则来拟合生成模型。然而，与训练单独的判别模型不同，生成模型本身被用来区分生成的数据与来自固定噪声分布的样本。由于NCE使用固定的噪声分布，当模型即使仅对观测变量的一小部分子集学习了近似正确的分布后，学习速度也会显著减慢。

最后，有些技术并不需要显式地定义一个概率分布，而是训练一个生成模型从期望的分布中采样。这种方法的好处是，这样的模型可以通过反向传播进行训练。最近这方面的工作包括生成随机网络（GSN）框架[5]，它扩展了广义去噪自编码器[4]：两者都可以看作是定义了一个参数化的马尔可夫链，即学习一个执行生成马尔可夫链一步的模型的参数。与GSNs相比，对抗网络框架在采样时不需要马尔可夫链。由于对抗网络在生成过程中不需要反馈循环，它们能更好地利用分段线性单元[19, 9, 10]，这提高了反向传播的性能，但在反馈循环中使用存在激活无界的问题。最近通过反向传播训练生成模型例子包括自编码变分贝叶斯[20]和随机反向传播[24]的研究。

## 3 对抗网络

对抗建模框架在模型均为多层感知机时应用最为直接。为了学习生成器在数据 $x$ 上的分布 $p_g$ ，我们首先定义输入噪声变量 $p_z(z)$ 的先验分布，然后将到数据空间的映射表示为 $G(z; \theta_g)$ ，其中 $G$ 是由参数为 $\theta_g$ 的多层感知机表示的可微函数。我们还定义了第二个多层感知机 $D(x; \theta_d)$ ，其输出为单个标量。 $D(x)$ 表示 $x$ 来自真实数据而非 $p_g$ 的概率。我们训练 $D$ 以最大化对训练样本和来自 $G$ 的样本分配正确标签的概率。同时训练 $G$ 以最小化对数损失 $\log(1 - D(G(z)))$ :

In other words,  $D$  and  $G$  play the following two-player minimax game with value function  $V(G, D)$ :

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (1)$$

In the next section, we present a theoretical analysis of adversarial nets, essentially showing that the training criterion allows one to recover the data generating distribution as  $G$  and  $D$  are given enough capacity, i.e., in the non-parametric limit. See Figure 1 for a less formal, more pedagogical explanation of the approach. In practice, we must implement the game using an iterative, numerical approach. Optimizing  $D$  to completion in the inner loop of training is computationally prohibitive, and on finite datasets would result in overfitting. Instead, we alternate between  $k$  steps of optimizing  $D$  and one step of optimizing  $G$ . This results in  $D$  being maintained near its optimal solution, so long as  $G$  changes slowly enough. This strategy is analogous to the way that SML/PCD [31, 29] training maintains samples from a Markov chain from one learning step to the next in order to avoid burning in a Markov chain as part of the inner loop of learning. The procedure is formally presented in Algorithm 1.

In practice, equation 1 may not provide sufficient gradient for  $G$  to learn well. Early in learning, when  $G$  is poor,  $D$  can reject samples with high confidence because they are clearly different from the training data. In this case,  $\log(1 - D(G(\mathbf{z})))$  saturates. Rather than training  $G$  to minimize  $\log(1 - D(G(\mathbf{z})))$  we can train  $G$  to maximize  $\log D(G(\mathbf{z}))$ . This objective function results in the same fixed point of the dynamics of  $G$  and  $D$  but provides much stronger gradients early in learning.

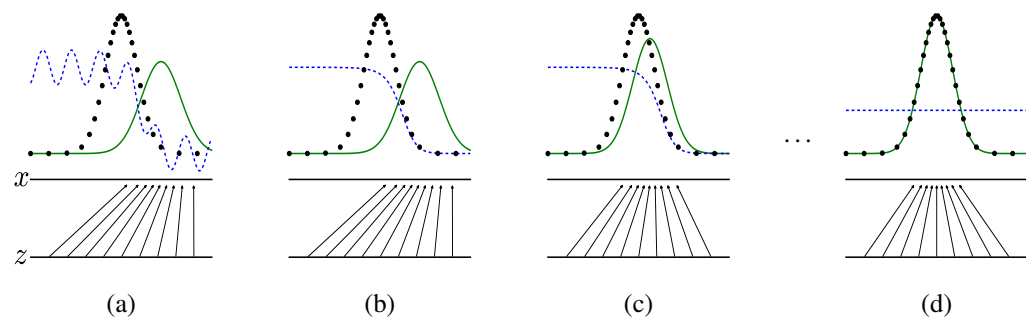


Figure 1: Generative adversarial nets are trained by simultaneously updating the discriminative distribution ( $D$ , blue, dashed line) so that it discriminates between samples from the data generating distribution (black, dotted line)  $p_{\mathbf{x}}$  from those of the generative distribution  $p_g$  ( $G$ ) (green, solid line). The lower horizontal line is the domain from which  $\mathbf{z}$  is sampled, in this case uniformly. The horizontal line above is part of the domain of  $\mathbf{x}$ . The upward arrows show how the mapping  $\mathbf{x} = G(\mathbf{z})$  imposes the non-uniform distribution  $p_g$  on transformed samples.  $G$  contracts in regions of high density and expands in regions of low density of  $p_g$ . (a) Consider an adversarial pair near convergence:  $p_g$  is similar to  $p_{\text{data}}$  and  $D$  is a partially accurate classifier. (b) In the inner loop of the algorithm  $D$  is trained to discriminate samples from data, converging to  $D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})}$ . (c) After an update to  $G$ , gradient of  $D$  has guided  $G(\mathbf{z})$  to flow to regions that are more likely to be classified as data. (d) After several steps of training, if  $G$  and  $D$  have enough capacity, they will reach a point at which both cannot improve because  $p_g = p_{\text{data}}$ . The discriminator is unable to differentiate between the two distributions, i.e.  $D(\mathbf{x}) = \frac{1}{2}$ .

## 4 Theoretical Results

The generator  $G$  implicitly defines a probability distribution  $p_g$  as the distribution of the samples  $G(\mathbf{z})$  obtained when  $\mathbf{z} \sim p_{\mathbf{z}}$ . Therefore, we would like Algorithm 1 to converge to a good estimator of  $p_{\text{data}}$ , if given enough capacity and training time. The results of this section are done in a non-parametric setting, e.g. we represent a model with infinite capacity by studying convergence in the space of probability density functions.

We will show in section 4.1 that this minimax game has a global optimum for  $p_g = p_{\text{data}}$ . We will then show in section 4.2 that Algorithm 1 optimizes Eq 1, thus obtaining the desired result.

In other words,  $D$  and  $G$  play the following two-player minimax game with value function  $V(G, D)$ :

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (1)$$

在下一节中，我们对对抗网络进行理论分析，本质上表明当 $G$ 和 $D$ 具有足够容量时（即在非参数极限下），训练准则允许恢复数据生成分布。关于该方法更通俗、更具教学性的解释请参见图1。在实践中，我们必须使用迭代的数值方法来实现这个博弈。在训练的内循环中将 $D$ 完全优化在计算上是不可行的，并且在有限数据集上会导致过拟合。因此，我们采用交替优化策略：先进行 $k$ 步 $D$ 的优化，再进行一步 $G$ 的优化。只要 $G$ 变化足够缓慢，这种策略就能使 $D$ 保持在其最优解附近。该方法类似于SML/PCD训练[31,29]中为避免在学习的内部循环中重新启动马尔可夫链，而将马尔可夫链的样本从一个学习步骤保持到下一个学习步骤的策略。该过程在算法1中正式给出。

在实践中，方程1可能无法为 $G$ 提供足够的学习梯度。在学习初期，当 $G$ 表现不佳时， $D$ 可能会以高置信度拒绝样本，因为这些样本与训练数据明显不同。这种情况下， $\log(1 - D(G(\mathbf{z})))$ 会达到饱和。与其训练 $G$ 以最小化 $\log(1 - D(G(\mathbf{z})))$ ，我们可以训练 $G$ 以最大化 $\log D(G(\mathbf{z}))$ 。该目标函数在 $G$ 和 $D$ 的动态过程中会产生相同的固定点，但在学习初期能提供更强的梯度。

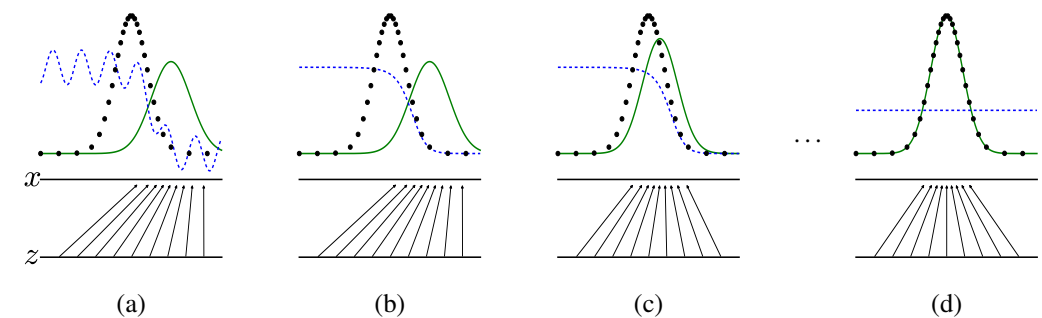


图1：生成对抗网络通过同时更新判别分布（ $D$ ，蓝色，虚线）进行训练，使其能够区分来自数据生成分布（黑色，点线） $p_{\mathbf{x}}$ 的样本与来自生成分布 $p_g$  ( $G$ ) (绿色，实线)的样本。下方水平线是 $\mathbf{z}$ 的采样域，在此例中为均匀分布。上方水平线是 $\mathbf{x}$ 域的一部分。向上箭头展示了映射 $\mathbf{x} = G(\mathbf{z})$ 如何在变换后的样本上施加非均匀分布 $p_g$ 。  $G$ 在高密度区域收缩，在 $p_g$ 的低密度区域扩展。(a) 考虑接近收敛的对抗对： $p_g$ 与 $p_{\text{data}}$ 相似，且 $D$ 是部分准确的分类器。(b) 在算法的内循环中， $D$ 被训练以区分数据样本，收敛至 $D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})}$ 。(c) 在 $G$ 更新后， $D$ 的梯度引导 $G(\mathbf{z})$ 流向更可能被分类为数据的区域。(d) 经过多步训练后，若 $G$ 和 $D$ 具备足够能力，它们将到达一个双方均无法改进的点，因为 $p_g = p_{\text{data}}$ 。此时判别器无法区分两个分布，即 $D(\mathbf{x}) = \frac{1}{2}$ 。

## 4 理论结果

生成器  $G$  隐式定义了一个概率分布  $p_g$ ，作为当  $\mathbf{z} \sim p_{\mathbf{z}}$  时获得的样本  $G(\mathbf{z})$  的分布。因此，如果给定足够的容量和训练时间，我们希望算法1能够收敛到  $p_{\text{data}}$  的一个良好估计量。本节的结果是在非参数设置下完成的，例如，我们通过研究概率密度函数空间中的收敛性来表示一个具有无限容量的模型。

我们将在4.1节中证明这个极小极大博弈对于  $p_g = p_{\text{data}}$  存在全局最优解。随后在4.2节中，我们将证明算法1能够优化方程1，从而获得期望的结果。

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**Algorithm 1** Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator,  $k$ , is a hyperparameter. We used  $k = 1$ , the least expensive option, in our experiments.

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**for** number of training iterations **do**

**for**  $k$  steps **do**

- Sample minibatch of  $m$  noise samples  $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$  from noise prior  $p_g(\mathbf{z})$ .
- Sample minibatch of  $m$  examples  $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$  from data generating distribution  $p_{\text{data}}(\mathbf{x})$ .
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[ \log D(\mathbf{x}^{(i)}) + \log \left( 1 - D(G(\mathbf{z}^{(i)})) \right) \right].$$

**end for**

- Sample minibatch of  $m$  noise samples  $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$  from noise prior  $p_g(\mathbf{z})$ .
- Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left( 1 - D(G(\mathbf{z}^{(i)})) \right).$$

**end for**

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

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#### 4.1 Global Optimality of $p_g = p_{\text{data}}$

We first consider the optimal discriminator  $D$  for any given generator  $G$ .

**Proposition 1.** *For  $G$  fixed, the optimal discriminator  $D$  is*

$$D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \quad (2)$$

*Proof.* The training criterion for the discriminator  $D$ , given any generator  $G$ , is to maximize the quantity  $V(G, D)$

$$\begin{aligned} V(G, D) &= \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) d\mathbf{x} + \int_{\mathbf{z}} p_{\mathbf{z}}(\mathbf{z}) \log(1 - D(G(\mathbf{z}))) d\mathbf{z} \\ &= \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) + p_g(\mathbf{x}) \log(1 - D(\mathbf{x})) d\mathbf{x} \end{aligned} \quad (3)$$

For any  $(a, b) \in \mathbb{R}^2 \setminus \{0, 0\}$ , the function  $y \rightarrow a \log(y) + b \log(1 - y)$  achieves its maximum in  $[0, 1]$  at  $\frac{a}{a+b}$ . The discriminator does not need to be defined outside of  $\text{Supp}(p_{\text{data}}) \cup \text{Supp}(p_g)$ , concluding the proof.  $\square$

Note that the training objective for  $D$  can be interpreted as maximizing the log-likelihood for estimating the conditional probability  $P(Y = y|\mathbf{x})$ , where  $Y$  indicates whether  $\mathbf{x}$  comes from  $p_{\text{data}}$  (with  $y = 1$ ) or from  $p_g$  (with  $y = 0$ ). The minimax game in Eq. 1 can now be reformulated as:

$$\begin{aligned} C(G) &= \max_D V(G, D) \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}} [\log(1 - D_G^*(G(\mathbf{z})))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[ \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[ \log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \end{aligned} \quad (4)$$

---

算法1 生成对抗网络的小批量随机梯度下降训练。应用于判别器的步骤数 $k$ 是一个超参数。在我们的实验中，我们使用了 $k = 1$ ，这是成本最低的选项。

---

或训练迭代次数为  $k$  步时执行

- 从噪声先验  $p_g(\mathbf{z})$  中采样包含  $m$  个噪声样本  $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$  的小批量样本。
- 从数据生成分布  $p_{\text{data}}(\mathbf{x})$  中抽取包含  $m$  个样本的小批量  $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(m)}\}$ 。
- 通过上升其随机梯度来更新判别器：

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[ \log D(\mathbf{x}^{(i)}) + \log \left( 1 - D(G(\mathbf{z}^{(i)})) \right) \right].$$

结束循环

- 从噪声先验  $p_g(\mathbf{z})$  中采样包含  $m$  个噪声样本  $\{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(m)}\}$  的小批量样本。
- 更新生成器，通过下降其随机梯度：

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left( 1 - D(G(\mathbf{z}^{(i)})) \right).$$

结束循环

基于梯度的更新可以使用任何标准的基于梯度的学习规则。我们在实验中使用了动量法。

---

#### 4.1 $p_g = p_{\text{data}}$ 的全局最优性

我们首先考虑对于任意给定的生成器  $G$  的最优判别器  $D$ 。

命题1. *For  $G$  fixed, the optimal discriminator  $D$  is*

$$D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \quad (2)$$

*Proof.* 给定任意生成器  $G$ ，判别器  $D$  的训练准则是最大化数量  $V(G, D)$

$$\begin{aligned} V(G, D) &= \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) d\mathbf{x} + \int_{\mathbf{z}} p_{\mathbf{z}}(\mathbf{z}) \log(1 - D(G(\mathbf{z}))) d\mathbf{z} \\ &= \int_{\mathbf{x}} p_{\text{data}}(\mathbf{x}) \log(D(\mathbf{x})) + p_g(\mathbf{x}) \log(1 - D(\mathbf{x})) d\mathbf{x} \end{aligned} \quad (3)$$

对于任意  $(a, b) \in \mathbb{R}^2 \setminus \{0, 0\}$ ，函数  $y \rightarrow a \log(y) + b \log(1 - y)$  在  $[0, 1]$  上于  $\frac{a}{a+b}$  处取得最大值。判别器无需在  $\text{Supp}(p_{\text{data}}) \cup \text{Supp}(p_g)$  之外定义，证明完毕。  $\square$

请注意， $D$  的训练目标可以解释为最大化估计条件概率  $P(Y = y|\mathbf{x})$  的对数似然，其中  $Y$  表示  $\mathbf{x}$  是来自  $p_{\text{data}}$  (对应  $y = 1$ ) 还是来自  $p_g$  (对应  $y = 0$ )。公式1中的极小极大博弈现在可以重新表述为：

$$\begin{aligned} C(G) &= \max_D V(G, D) \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}} [\log(1 - D_G^*(G(\mathbf{z})))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))] \\ &= \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[ \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[ \log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \end{aligned} \quad (4)$$

**Theorem 1.** *The global minimum of the virtual training criterion  $C(G)$  is achieved if and only if  $p_g = p_{data}$ . At that point,  $C(G)$  achieves the value  $-\log 4$ .*

*Proof.* For  $p_g = p_{data}$ ,  $D_G^*(\mathbf{x}) = \frac{1}{2}$ , (consider Eq. 2). Hence, by inspecting Eq. 4 at  $D_G^*(\mathbf{x}) = \frac{1}{2}$ , we find  $C(G) = \log \frac{1}{2} + \log \frac{1}{2} = -\log 4$ . To see that this is the best possible value of  $C(G)$ , reached only for  $p_g = p_{data}$ , observe that

$$\mathbb{E}_{\mathbf{x} \sim p_{data}} [-\log 2] + \mathbb{E}_{\mathbf{x} \sim p_g} [-\log 2] = -\log 4$$

and that by subtracting this expression from  $C(G) = V(D_G^*, G)$ , we obtain:

$$C(G) = -\log(4) + KL\left(p_{data} \left\| \frac{p_{data} + p_g}{2} \right\| \right) + KL\left(p_g \left\| \frac{p_{data} + p_g}{2} \right\| \right) \quad (5)$$

where KL is the Kullback–Leibler divergence. We recognize in the previous expression the Jensen–Shannon divergence between the model’s distribution and the data generating process:

$$C(G) = -\log(4) + 2 \cdot JSD(p_{data} \| p_g) \quad (6)$$

Since the Jensen–Shannon divergence between two distributions is always non-negative and zero only when they are equal, we have shown that  $C^* = -\log(4)$  is the global minimum of  $C(G)$  and that the only solution is  $p_g = p_{data}$ , i.e., the generative model perfectly replicating the data generating process.  $\square$

## 4.2 Convergence of Algorithm 1

**Proposition 2.** *If  $G$  and  $D$  have enough capacity, and at each step of Algorithm 1, the discriminator is allowed to reach its optimum given  $G$ , and  $p_g$  is updated so as to improve the criterion*

$$\mathbb{E}_{\mathbf{x} \sim p_{data}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))]$$

*then  $p_g$  converges to  $p_{data}$*

*Proof.* Consider  $V(G, D) = U(p_g, D)$  as a function of  $p_g$  as done in the above criterion. Note that  $U(p_g, D)$  is convex in  $p_g$ . The subderivatives of a supremum of convex functions include the derivative of the function at the point where the maximum is attained. In other words, if  $f(x) = \sup_{\alpha \in \mathcal{A}} f_\alpha(x)$  and  $f_\alpha(x)$  is convex in  $x$  for every  $\alpha$ , then  $\partial f_\beta(x) \in \partial f$  if  $\beta = \arg \sup_{\alpha \in \mathcal{A}} f_\alpha(x)$ . This is equivalent to computing a gradient descent update for  $p_g$  at the optimal  $D$  given the corresponding  $G$ .  $\sup_D U(p_g, D)$  is convex in  $p_g$  with a unique global optima as proven in Thm 1, therefore with sufficiently small updates of  $p_g$ ,  $p_g$  converges to  $p_x$ , concluding the proof.  $\square$

In practice, adversarial nets represent a limited family of  $p_g$  distributions via the function  $G(\mathbf{z}; \theta_g)$ , and we optimize  $\theta_g$  rather than  $p_g$  itself. Using a multilayer perceptron to define  $G$  introduces multiple critical points in parameter space. However, the excellent performance of multilayer perceptrons in practice suggests that they are a reasonable model to use despite their lack of theoretical guarantees.

## 5 Experiments

We trained adversarial nets on a range of datasets including MNIST[23], the Toronto Face Database (TFD) [28], and CIFAR-10 [21]. The generator nets used a mixture of rectifier linear activations [19, 9] and sigmoid activations, while the discriminator net used maxout [10] activations. Dropout [17] was applied in training the discriminator net. While our theoretical framework permits the use of dropout and other noise at intermediate layers of the generator, we used noise as the input to only the bottommost layer of the generator network.

We estimate probability of the test set data under  $p_g$  by fitting a Gaussian Parzen window to the samples generated with  $G$  and reporting the log-likelihood under this distribution. The  $\sigma$  parameter

定理 1. *The global minimum of the virtual training criterion  $C(G)$  is achieved if and only if  $p_g = p_{data}$ . At that point,  $C(G)$  achieves the value  $-\log 4$ .*

*Proof.* 对于  $p_g = p_{data}$ ,  $D_G^*(\mathbf{x}) = \frac{1}{2}$  (考虑公式2)。因此, 通过检查公式4在  $D_G^*(\mathbf{x}) = \frac{1}{2}$  处的情况, 我们发现  $C(G) = \log \frac{1}{2} + \log \frac{1}{2} = -\log 4$ 。为了说明这是  $C(G)$  可能达到的最佳值, 且仅当  $p_g = p_{data}$  时实现, 可以观察到

$$\mathbb{E}_{\mathbf{x} \sim p_{data}} [-\log 2] + \mathbb{E}_{\mathbf{x} \sim p_g} [-\log 2] = -\log 4$$

并且通过从  $C(G) = V(D_G^*, G)$  中减去这个表达式, 我们得到:

$$C(G) = -\log(4) + KL\left(p_{data} \left\| \frac{p_{data} + p_g}{2} \right\| \right) + KL\left(p_g \left\| \frac{p_{data} + p_g}{2} \right\| \right) \quad (5)$$

其中KL是Kullback–Leibler散度。我们在上述表达式中识别出模型分布与数据生成过程之间的Jensen–Shannon散度:

$$C(G) = -\log(4) + 2 \cdot JSD(p_{data} \| p_g) \quad (6)$$

由于两个分布之间的Jensen-Shannon散度总是非负的, 且仅当它们相等时为零, 我们已经证明  $C^* = -\log(4)$  是  $C(G)$  的全局最小值, 并且唯一解是  $p_g = p_{data}$ , 即生成模型完美复现了数据生成过程。  $\square$

## 4.2 算法1的收敛性

命题2. *If  $G$  and  $D$  have enough capacity, and at each step of Algorithm 1, the discriminator is allowed to reach its optimum given  $G$ , and  $p_g$  is updated so as to improve the criterion*

$$\mathbb{E}_{\mathbf{x} \sim p_{data}} [\log D_G^*(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_g} [\log(1 - D_G^*(\mathbf{x}))]$$

*then  $p_g$  converges to  $p_{data}$*

*Proof.* 将上述准则中的  $V(G, D) = U(p_g, D)$  视为  $p_g$  的函数。注意  $U(p_g, D)$  在  $p_g$  中是凸的。凸函数上确界的次导数包含在达到最大值点处函数的导数。换言之, 若  $f(x) = \sup_{\alpha \in \mathcal{A}} f_\alpha(x)$  且对每个  $\alpha$ ,  $f_\alpha(x)$  在  $x$  中为凸, 则当  $\beta = \arg \sup_{\alpha \in \mathcal{A}} f_\alpha(x)$  时, 有  $\partial f_\beta(x) \in \partial f$ 。这等价于在给定对应  $G$  的情况下, 对最优  $D$  处的  $p_g$  计算梯度下降更新。如定理1所证,  $\sup_D U(p_g, D)$  在  $p_g$  中为凸且具有唯一全局最优解, 因此当  $p_g$  的更新足够小时,  $p_g$  收敛于  $p_x$ , 证明完成。  $\square$

在实践中, 对抗网络通过函数  $G(\mathbf{z})$  表示一个有限的  $p_g$  分布族;  $\theta_g$ , 我们优化的是  $\theta_g$  而非  $p_g$  本身。使用多层感知器定义  $G$  会在参数空间中引入多个临界点。然而, 多层感知器在实践中表现优异, 这表明尽管缺乏理论保证, 它们仍是一个合理的模型选择。

## 5 实验

我们在包括MNIST[23]、多伦多人脸数据库(TFD)[28]和CIFAR-10[21]在内的多个数据集上训练了对抗网络。生成器网络混合使用了线性整流激活函数[19,9]和sigmoid激活函数, 而判别器网络则采用了maxout[10]激活函数。在训练判别器网络时应用了Dropout[17]技术。虽然我们的理论框架允许在生成器的中间层使用Dropout和其他噪声, 但我们仅将噪声作为输入应用于生成器网络的最底层。

我们通过将高斯Parzen窗口拟合到使用  $G$  生成的样本上, 并报告在此分布下的对数似然, 来估计测试集数据在  $p_g$  下的概率。 $\sigma$  参数



| Model            | MNIST                         | TFD                             |
|------------------|-------------------------------|---------------------------------|
| DBN [3]          | 138 $\pm$ 2                   | 1909 $\pm$ 66                   |
| Stacked CAE [3]  | 121 $\pm$ 1.6                 | <b>2110 <math>\pm</math> 50</b> |
| Deep GSN [6]     | 214 $\pm$ 1.1                 | 1890 $\pm$ 29                   |
| Adversarial nets | <b>225 <math>\pm</math> 2</b> | <b>2057 <math>\pm</math> 26</b> |

Table 1: Parzen window-based log-likelihood estimates. The reported numbers on MNIST are the mean log-likelihood of samples on test set, with the standard error of the mean computed across examples. On TFD, we computed the standard error across folds of the dataset, with a different  $\sigma$  chosen using the validation set of each fold. On TFD,  $\sigma$  was cross validated on each fold and mean log-likelihood on each fold were computed. For MNIST we compare against other models of the real-valued (rather than binary) version of dataset.

of the Gaussians was obtained by cross validation on the validation set. This procedure was introduced in Breuleux *et al.* [8] and used for various generative models for which the exact likelihood is not tractable [25, 3, 5]. Results are reported in Table 1. This method of estimating the likelihood has somewhat high variance and does not perform well in high dimensional spaces but it is the best method available to our knowledge. Advances in generative models that can sample but not estimate likelihood directly motivate further research into how to evaluate such models.

In Figures 2 and 3 we show samples drawn from the generator net after training. While we make no claim that these samples are better than samples generated by existing methods, we believe that these samples are at least competitive with the better generative models in the literature and highlight the potential of the adversarial framework.

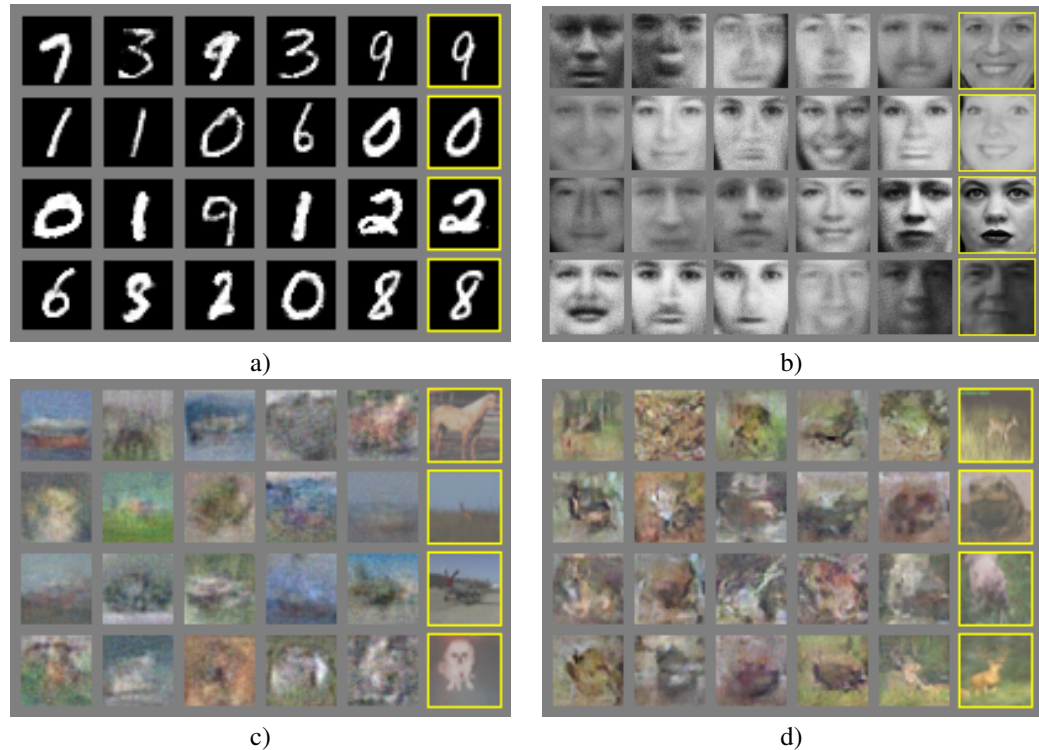


Figure 2: Visualization of samples from the model. Rightmost column shows the nearest training example of the neighboring sample, in order to demonstrate that the model has not memorized the training set. Samples are fair random draws, not cherry-picked. Unlike most other visualizations of deep generative models, these images show actual samples from the model distributions, not conditional means given samples of hidden units. Moreover, these samples are uncorrelated because the sampling process does not depend on Markov chain mixing. a) MNIST b) TFD c) CIFAR-10 (fully connected model) d) CIFAR-10 (convolutional discriminator and “deconvolutional” generator)

| Model            | MNIST                         | TFD                             |
|------------------|-------------------------------|---------------------------------|
| DBN [3]          | 138 $\pm$ 2                   | 1909 $\pm$ 66                   |
| Stacked CAE [3]  | 121 $\pm$ 1.6                 | <b>2110 <math>\pm</math> 50</b> |
| Deep GSN [6]     | 214 $\pm$ 1.1                 | 1890 $\pm$ 29                   |
| Adversarial nets | <b>225 <math>\pm</math> 2</b> | <b>2057 <math>\pm</math> 26</b> |

表1: 基于Parzen窗口的对数似然估计。在MNIST上报告的数字是测试集样本的平均对数似然, 均值标准误差通过样本计算得出。在TFD上, 我们计算了数据集各折的标准误差, 其中 $\sigma$ 是通过每折的验证集选择的。在TFD上,  $\sigma$ 在每折上进行了交叉验证, 并计算了每折的平均对数似然。对于MNIST, 我们与数据集的实值版本(而非二值版本)的其他模型进行了比较。

高斯函数的参数是通过在验证集上进行交叉验证获得的。这一方法由Breuleux *et al.* [8]提出, 并已应用于多种精确似然难以处理的生成模型[25, 3, 5]。结果呈现在表1中。这种似然估计方法方差较大, 在高维空间中表现不佳, 但据我们所知它仍是当前最优方法。生成模型的发展使得采样成为可能而直接估计似然仍具挑战, 这直接推动了对如何评估此类模型的进一步研究。

在图2和图3中, 我们展示了训练后从生成器网络中抽取的样本。虽然我们并不声称这些样本优于现有方法生成的样本, 但我们认为这些样本至少与文献中较好的生成模型相媲美, 并突显了对抗性框架的潜力。

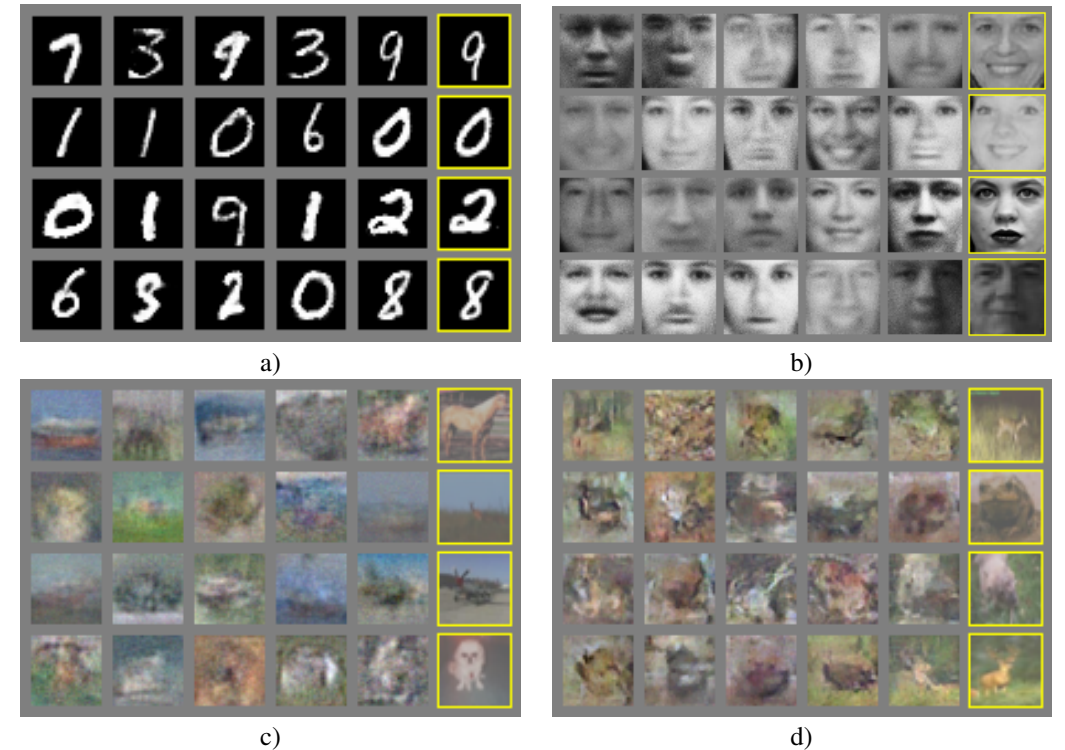


图2: 模型样本的可视化。最右侧一列展示了相邻样本的最近训练示例, 以证明模型并未记忆训练集。样本均为公平随机抽取, 非精心挑选。与大多数深度生成模型的可视化不同, 这些图像展示的是模型分布中的实际样本, 而非给定隐藏单元样本的条件均值。此外, 这些样本互不相关, 因为采样过程不依赖于马尔可夫链混合。a) MNIST b) TFD c) CIFAR-10 (全连接模型) d) CIFAR-10 (卷积判别器与“反卷积”生成器)



Figure 3: Digits obtained by linearly interpolating between coordinates in  $z$  space of the full model.

|                   | Deep directed graphical models             | Deep undirected graphical models                                                          | Generative autoencoders                                                        | Adversarial models                                                             |
|-------------------|--------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Training          | Inference needed during training.          | Inference needed during training. MCMC needed to approximate partition function gradient. | Enforced tradeoff between mixing and power of reconstruction generation        | Synchronizing the discriminator with the generator. Helvetica.                 |
| Inference         | Learned approximate inference              | Variational inference                                                                     | MCMC-based inference                                                           | Learned approximate inference                                                  |
| Sampling          | No difficulties                            | Requires Markov chain                                                                     | Requires Markov chain                                                          | No difficulties                                                                |
| Evaluating $p(x)$ | Intractable, may be approximated with AIS  | Intractable, may be approximated with AIS                                                 | Not explicitly represented, may be approximated with Parzen density estimation | Not explicitly represented, may be approximated with Parzen density estimation |
| Model design      | Nearly all models incur extreme difficulty | Careful design needed to ensure multiple properties                                       | Any differentiable function is theoretically permitted                         | Any differentiable function is theoretically permitted                         |

Table 2: Challenges in generative modeling: a summary of the difficulties encountered by different approaches to deep generative modeling for each of the major operations involving a model.

## 6 Advantages and disadvantages

This new framework comes with advantages and disadvantages relative to previous modeling frameworks. The disadvantages are primarily that there is no explicit representation of  $p_g(x)$ , and that  $D$  must be synchronized well with  $G$  during training (in particular,  $G$  must not be trained too much without updating  $D$ , in order to avoid “the Helvetica scenario” in which  $G$  collapses too many values of  $z$  to the same value of  $x$  to have enough diversity to model  $p_{\text{data}}$ ), much as the negative chains of a Boltzmann machine must be kept up to date between learning steps. The advantages are that Markov chains are never needed, only backprop is used to obtain gradients, no inference is needed during learning, and a wide variety of functions can be incorporated into the model. Table 2 summarizes the comparison of generative adversarial nets with other generative modeling approaches.

The aforementioned advantages are primarily computational. Adversarial models may also gain some statistical advantage from the generator network not being updated directly with data examples, but only with gradients flowing through the discriminator. This means that components of the input are not copied directly into the generator’s parameters. Another advantage of adversarial networks is that they can represent very sharp, even degenerate distributions, while methods based on Markov chains require that the distribution be somewhat blurry in order for the chains to be able to mix between modes.

## 7 Conclusions and future work

This framework admits many straightforward extensions:

1. A *conditional generative* model  $p(x | c)$  can be obtained by adding  $c$  as input to both  $G$  and  $D$ .
2. *Learned approximate inference* can be performed by training an auxiliary network to predict  $z$  given  $x$ . This is similar to the inference net trained by the wake-sleep algorithm [15] but with the advantage that the inference net may be trained for a fixed generator net after the generator net has finished training.



图3：通过在完整模型的 $z$ 空间中对坐标进行线性插值获得的数字。

|                   | Deep directed graphical models             | Deep undirected graphical models                                                          | Generative autoencoders                                                        | Adversarial models                                                             |
|-------------------|--------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Training          | Inference needed during training.          | Inference needed during training. MCMC needed to approximate partition function gradient. | Enforced tradeoff between mixing and power of reconstruction generation        | Synchronizing the discriminator with the generator. Helvetica.                 |
| Inference         | Learned approximate inference              | Variational inference                                                                     | MCMC-based inference                                                           | Learned approximate inference                                                  |
| Sampling          | No difficulties                            | Requires Markov chain                                                                     | Requires Markov chain                                                          | No difficulties                                                                |
| Evaluating $p(x)$ | Intractable, may be approximated with AIS  | Intractable, may be approximated with AIS                                                 | Not explicitly represented, may be approximated with Parzen density estimation | Not explicitly represented, may be approximated with Parzen density estimation |
| Model design      | Nearly all models incur extreme difficulty | Careful design needed to ensure multiple properties                                       | Any differentiable function is theoretically permitted                         | Any differentiable function is theoretically permitted                         |

表2：生成建模中的挑战：针对涉及模型的主要操作，总结了不同深度生成建模方法遇到的困难。

## 6 优点和缺点

这一新框架相较于以往的建模框架，既有优势也有不足。不足之处主要在于没有对 $p_g(x)$ 的显式表示，且训练过程中 $D$ 必须与 $G$ 良好同步（具体而言，为避免“Helvetica情景”——即 $G$ 将过多 $z$ 值坍缩到同一 $x$ 值，导致缺乏足够多样性来建模 $p_{\text{data}}$ ——必须确保在更新 $D$ 前不对 $G$ 进行过度训练），这类似于玻尔兹曼机的负链必须在学习步骤间保持更新。其优势则在于完全不需要马尔可夫链，仅通过反向传播获取梯度，学习过程中无需推断，且模型能兼容多种函数。表2总结了生成对抗网络与其他生成建模方法的对比。

上述优势主要是计算上的。对抗模型还可能从生成器网络不直接使用数据样本更新，而仅通过判别器传递的梯度进行更新中获得一些统计优势。这意味着输入的组成部分不会直接复制到生成器的参数中。对抗网络的另一个优势是，它们能够表示非常尖锐、甚至退化的分布，而基于马尔可夫链的方法要求分布必须有一定程度的模糊性，以便链能够在不同模式之间混合。

## 7 结论与未来工作

该框架允许许多直接的扩展：

1. 通过向 $G$ 和 $D$ 同时添加 $c$ 作为输入，可以获得一个*conditional generative*模型 $p(x | c)$ 。
2. 可以通过训练一个辅助网络，在给定 $x$ 的情况下预测 $z$ 来执行*Learned approximate inference*。这类似于由唤醒-睡眠算法[15]训练的推断网络，但其优势在于推断网络可以在生成器网络完成训练后，针对固定的生成器网络进行训练。

3. One can approximately model all conditionals  $p(\boldsymbol{x}_S \mid \boldsymbol{x}_S)$  where  $S$  is a subset of the indices of  $\boldsymbol{x}$  by training a family of conditional models that share parameters. Essentially, one can use adversarial nets to implement a stochastic extension of the deterministic MP-DBM [11].
4. *Semi-supervised learning*: features from the discriminator or inference net could improve performance of classifiers when limited labeled data is available.
5. *Efficiency improvements*: training could be accelerated greatly by devising better methods for coordinating  $G$  and  $D$  or determining better distributions to sample  $\mathbf{z}$  from during training.

This paper has demonstrated the viability of the adversarial modeling framework, suggesting that these research directions could prove useful.

### Acknowledgments

We would like to acknowledge Patrice Marcotte, Olivier Delalleau, Kyunghyun Cho, Guillaume Alain and Jason Yosinski for helpful discussions. Yann Dauphin shared his Parzen window evaluation code with us. We would like to thank the developers of Pylearn2 [12] and Theano [7, 1], particularly Frédéric Bastien who rushed a Theano feature specifically to benefit this project. Arnaud Bergeron provided much-needed support with L<sup>A</sup>T<sub>E</sub>X typesetting. We would also like to thank CIFAR, and Canada Research Chairs for funding, and Compute Canada, and Calcul Québec for providing computational resources. Ian Goodfellow is supported by the 2013 Google Fellowship in Deep Learning. Finally, we would like to thank Les Trois Brasseurs for stimulating our creativity.

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3. 通过训练一组共享参数的条件模型，可以近似建模所有条件分布 $p(\boldsymbol{x}_S \mid \boldsymbol{x}_S)$ ，其中 $S$ 是 $\boldsymbol{x}$ 索引的子集。本质上，可以利用对抗网络实现确定性MP-DBM[11]的随机扩展。
4. *Semi-supervised learning*：当标记数据有限时，判别器或推断网络提取的特征可提升分类器性能。
5. 通过设计更优的 $G$ 与 $D$ 协调方法，或确定训练过程中更佳的 $\mathbf{z}$ 采样分布，可大幅加速*Efficiency improvements*:训练。

本文论证了对抗建模框架的可行性，表明这些研究方向可能具有实用价值。

### 致谢

我们要感谢Patrice Marcotte、Olivier Delalleau、Kyunghyun Cho、Guillaume Alain和Jason Yosinski的有益讨论。Yann Dauphin与我们分享了他的Parzen窗口评估代码。我们还要感谢Pylearn2 [12]和Theano [7, 1]的开发者，特别是Frédéric Bastien，他专门为这个项目紧急添加了一项Theano功能。Arnaud Bergeron在L<sup>A</sup>T<sub>E</sub>X排版方面提供了急需的支持。我们还要感谢CIFAR和加拿大研究主席项目提供的资金支持，以及Compute Canada和Calcul Québec提供的计算资源。Ian Goodfellow获得了2013年谷歌深度学习奖学金的支持。最后，我们要感谢Les Trois Brasseurs激发了我们的创造力。

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